From the candidate universe to CTAs

OncoRef 1.31.0 | 5 November 2026

Research candidate selection, evidence status and verified sequence use.

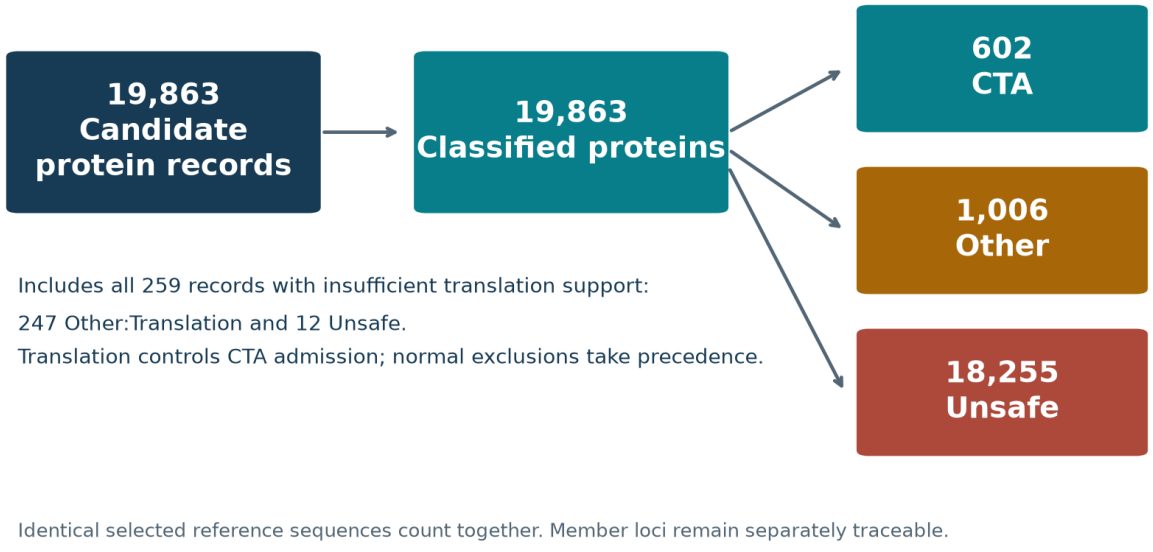
What OncoRef calls a CTA

An OncoRef CTA candidate is a reference-resolved protein with supported cancer expression and qualifying reproductive or developmental association that passes the named normal-tissue evidence, coverage and selection rules. CTA admission requires qualifying coding annotation or supported ncORF evidence and qualification of the exact selected translation. Every input record is assessed, including those with insufficient translation support.

The current human policy produces 602 CTA, 18,255 Unsafe and 1,006 Other proteins. It evaluates 19,863 candidate proteins in the complete inventory, counting identical selected reference sequences together and retaining every member locus. This is not a complete inventory of splice isoforms or modified proteoforms.

CTA is a research candidate designation; it does not establish peptide-HLA presentation, receptor recognition, therapeutic abundance or clinical safety. Reproductive/developmental support includes adult reproductive tissues, placenta, extraembryonic tissues, embryo and fetus. These routes remain distinguishable; developmental association does not imply adult testis restriction.

Every candidate record receives a classification



Identical selected reference sequences count together. Member loci remain separately traceable.

From candidates to CTAs

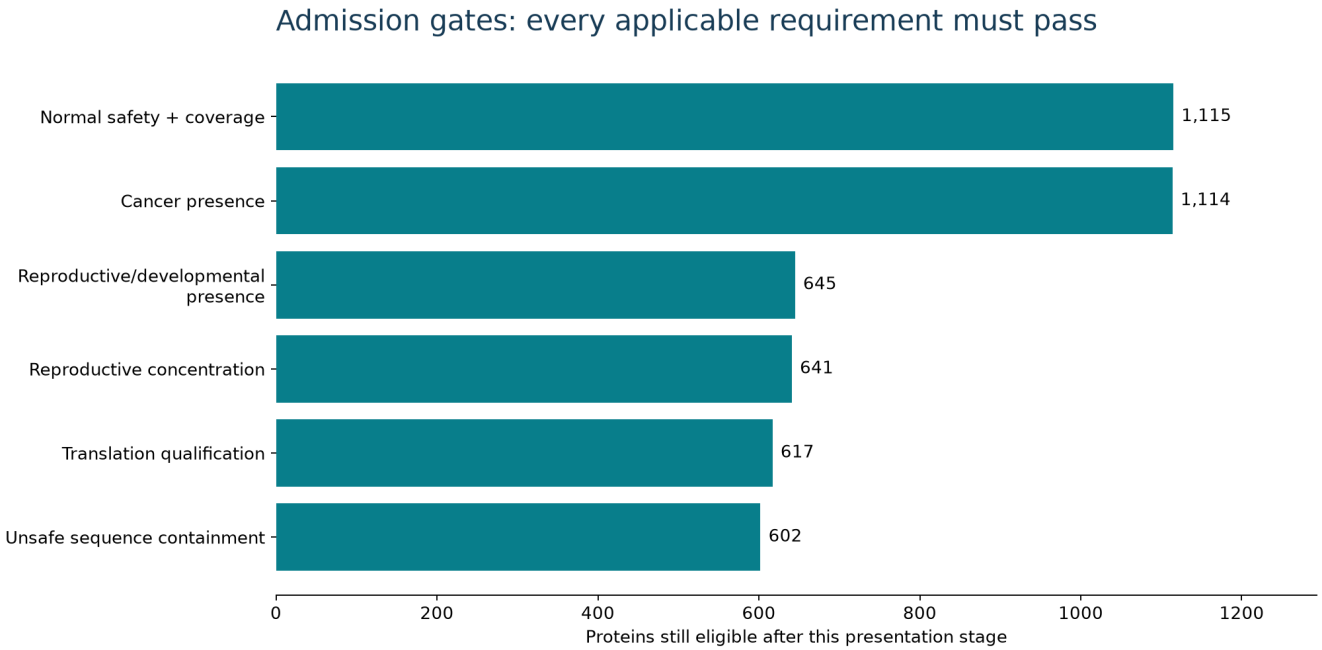
All 19,863 candidate records enter the same normal-evidence and admission path. The 259 records formerly removed at entry stay visible: 254 have only NMD/CDS-not-defined reference transcripts and 5 have only single-study Ribo-seq support. Insufficient translation support does not establish absence or normal-tissue safety.

CTA admission requires qualifying translation support and the separately qualified exact reference record. Applicable normal exclusion takes precedence and establishes Unsafe; otherwise insufficient translation qualification yields Other:Translation and sequence review. An annotated coding sequence is not experimental proof of protein production.

Apply all normal-tissue evidence independently. Accepted exclusion evidence establishes Unsafe. Unresolved findings or missing required coverage prevent admission and require review; they do not establish Unsafe. Every target must also pass reproductive/developmental association, cancer presence, concentration, translation qualification and sequence-containment rules.

The chart uses cumulative counts. Of 1,115 proteins passing normal assessment, cancer presence retains 1,114; reproductive/developmental presence then narrows these to 645. The previous order showed 645 at both gates because every reproductive/developmental survivor already met cancer presence. One compatible positive tumor RNA sample or reviewed tumor protein makes cancer presence a broad gate, not an abundance or prevalence requirement.

Every applicable evidence layer is still evaluated, including after an early exclusion. Unsafe takes precedence. Display order changes neither final classifications nor the priority used to assign Other subtypes.



Published nominations through the funnel

All 15 registered nomination papers appear in the companion funnel. Their union contains 2,514 proteins: 480 finish as CTA, 1,876 as Unsafe and 158 as Other. A protein can occur in several papers, so individual paper rows must not be summed.

The 17,349 candidates without a registered-paper annotation undergo the same assessment and contribute 122 CTAs. Prior nomination is recorded provenance, not a required admission gate. Absence from this finite registry does not establish novelty or absence from the literature.

A publication's measured evidence can support reproductive/developmental association, tumor expression, translation or normal-tissue counterevidence when it meets the corresponding evidence rule. A list label alone does not establish those facts. Germ-cell, testis-biased, placental and tumor-vascular lists retain their original scope alongside CTA nominations.

Every registered nomination paper follows the same admission gates

	Input	Normal assessment	Cancer presence	Repro. / dev.	Concentration	Translation	Final CTA
da Silva et al. (2017)	1,017	512	512	432	432	420	411
Wang et al. (2016)	988	292	292	279	279	271	269
Chang et al. (2019)	985	292	292	289	289	281	281
Gong et al. (2021)	729	370	370	357	356	349	346
Bruggeman et al. (2018)	692	12	12	12	12	12	12
Jamin et al. (2021)	469	39	39	38	38	36	33
Loriot et al. (2025)	202	86	86	86	86	84	83
Carter et al. (2023)	103	54	54	54	54	53	53
Hirbo et al. (2015)	78	10	10	10	10	10	8
Seager et al. (2024)	15	11	11	11	11	10	10
Sasaroli et al. (2011)	13	0	0	0	0	0	0
Bradley et al. (2020)	9	0	0	0	0	0	0
Garg et al. (2007)	5	3	3	3	3	3	3
Boël et al. (1995)	2	1	1	1	1	1	1
Bai et al. (2016)	1	0	0	0	0	0	0
Registered-paper union	2,514	592	592	505	504	490	480
No registered-paper annotation	17,349	523	522	140	137	127	122
Complete inventory	19,863	1,115	1,114	645	641	617	602

Counts are cumulative. Color shows the fraction of each row's input remaining.
Paper rows overlap and must not be added. The union and unannotated rows partition the full inventory.
Nomination scopes include CTA, germ-cell, placental and vascular lists. Membership records provenance; measured evidence determines admission.

Download every paper-by-gate count and its source scope

Registered nomination papers and extraction scope

da Silva et al. (2017): At least 90% normal-tissue RNA attributable to testis; tumor positivity not required; Gene-labelled tumor proteomics; unique-locus peptide assignment is not established; RSEM >1 in at least 10% of samples in a tumor type; RNA nomination

Wang et al. (2016): Cancer-associated RNA nomination; no antigen validation asserted

Chang et al. (2019): Testis-specific coding gene expressed in at least 1% of TGCT; no broad somatic-cancer claim

Gong et al. (2021): Normal reproductive-tissue enrichment; no tumor validation; Placental RNA enrichment; no tumor-antigen validation

Bruggeman et al. (2018): GC RNA nomination across cancer types; no antigen validation asserted; GC RNA nomination; inclusion depends on TGCT under this study's thresholds

Jamin et al. (2021): CT-pattern microarray nomination; tumor upregulation with no corresponding normal-tissue detection; Core CT annotation; underlying microarray probes may recognize several loci

Loriot et al. (2025): Cancer-germline RNA nomination; strict and preferential classes remain distinct

Carter et al. (2023): Final CT-expression set without detectable thymic expression; functional vaccination evidence is not assigned to every gene

Hirbo et al. (2015): HPA v13 placenta-enriched nominations; 30/86 fast-evolving subset retained as metadata; not tumor validation

Seager et al. (2024): Member of the published 17-target CTA profiling panel; includes differentiation antigen MLANA

Sasaroli et al. (2011): Tumor-vascular candidate list; includes secreted and transmembrane proteins; not a CTA-restricted list

Bradley et al. (2020): CPA expression nominations; HLA peptide and antigen-specific T-cell validation for VGLL1

Garg et al. (2007): Ovarian-cancer RNA/protein expression and patient antibody responses; no HLA presentation or T-cell validation claimed

Boël et al. (1995): Original BAGE tumor-antigen nomination and autologous CTL recognition; contemporary paralog resolution is not inferred

Bai et al. (2016): Ovarian-tumor and vascular expression/function; no normal-tissue exclusivity claim

Normal-tissue consequences and expression evidence

Vital, Protected, Manageable and Disposable describe the consequence and tolerability of injury to a specified tissue/cell compartment. They are qualitative research priorities, not measured fatality probabilities. Immediate function, regenerative reserve, persistent depletion, anatomy and inflammatory amplification all matter.

The central registry distinguishes 39 bulk tissues and 68 cell populations, with source-linked rationale. A cell-level exception requires resolved tissue and cell identity. An unresolved or mixed label cannot earn a narrower allowance; reproductive/developmental membership is a separate anatomical axis.

RNA detection, accessible surface protein, peptide-HLA presentation, functional recognition and demonstrated injury are distinct evidence levels. CAR and TCR target availability are different questions. A weak initiating target can still produce inflammatory injury in other cells. Missing evidence is never measured zero.

Vital: Highest protection priority: plausible life-threatening failure, major irreversible neurological/sensory loss, or loss of indispensable regenerative support.

Heart and alveolar injury: direct human T-cell injury and animal depletion evidence support high-consequence concerns; neither estimates a universal fatality rate.

Cardiovascular toxicity and titin cross-reactivity of affinity-enhanced T cells in myeloma and melanoma; Targeted Type 2 Alveolar Cell Depletion: A Dynamic Functional Model for Lung Injury Repair

Protected: Persistent functional or structural injury is a major concern; compensation is incomplete or direct immune-depletion evidence is insufficient.

Persistent salivary injury and impaired ciliary function illustrate lasting functional consequences. Disease and depletion models are not interchangeable with immune killing.

Sicca Syndrome Associated with Immune Checkpoint Inhibitor Therapy; Lung function from school age to adulthood in primary ciliary dyskinesia; Parent stem cells can serve as niches for their daughter cells

Manageable: A defined compartment has credible replacement, compensation or supportive-treatment precedent; acute inflammation and persistent recognition still require review.

B-cell depletion and thyroid injury have clinical replacement or supportive-treatment precedent. Management remains conditional on the exact compartment and treatment.

Tisagenlecleucel in Children and Young Adults with B-Cell Lymphoblastic Leukemia; Immune-Related Thyroiditis with Immune Checkpoint Inhibitors

Disposable: Narrow conditional allowance for loss of a specified compartment; does not imply painless, reversible or inflammation-free ablation.

Gallbladder removal provides a narrow surgical precedent. Melanocyte-targeted T cells also damaged eye and ear tissues: a cutaneous allowance cannot extend to every melanocyte compartment.

Treatment for Gallstones; Gene therapy with human and mouse T-cell receptors mediates cancer regression and targets normal tissues expressing cognate antigen

Central tissue registry, compartment rationales and primary references

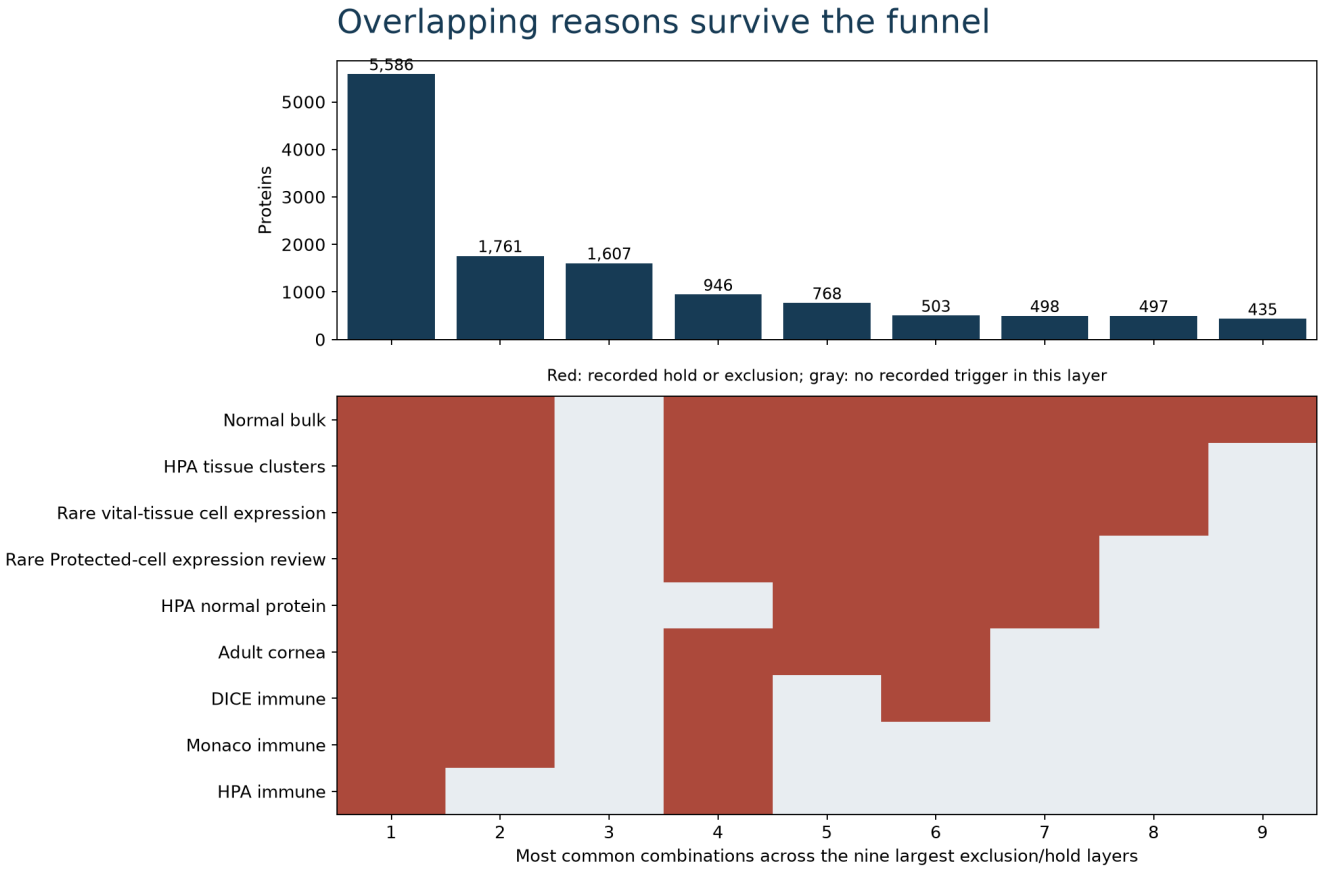
Evidence has multiple independent meanings

Review completion, interpretation and policy effect are separate. A completed review can retain an unresolved interpretation. A rejected protein claim remains visible with its rationale; rejecting that claim does not prove normal-tissue absence.

All 196 historical findings have linked current dispositions. MAGEC2, LGALS13 and PLCZ1 retain material holds. Their evidence access depth, reagent identity, adult/fetal and isoform limits are stated explicitly. The broader literature review remains incomplete.

Fatal neurological injury in the MAGE-A3-reactive TCR cohort remains product-specific evidence. The MAGEA12 and EPS8L2 explanations are both retained as unresolved. MAGEA12 remains Unsafe under an independent rare-cell finding. Neither receptor causality nor favorable clinical credit transfers across a gene family.

Morgan et al. 2013: neurological injury report · Martin et al. 2021: MAGEA12/EPS8L2 re-examination



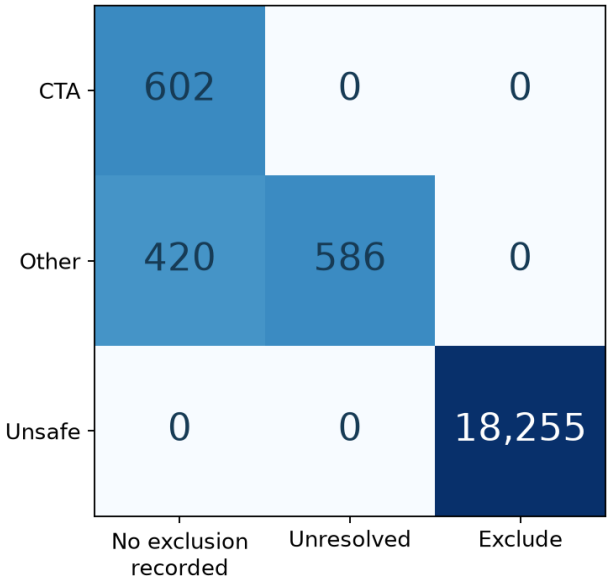
Translation, identity and downstream sequence use

Translation qualification asks whether the exact selected primary-locus record supports target sequence use. Complete coding annotation, start/stop codons, biotype and sequence integrity are assessed separately from normal-tissue hazard. An alternate-locus protein cannot silently rescue an incomplete primary record.

The verified bundle retains original sequences, every source occurrence, qualification reasons and coverage gaps. It supplies contiguous AA20 segments with original coordinates and never joins across ambiguity, internal stops or different records. Selenocysteine-containing records require review under this AA20 contract; this is not a claim that the protein is unexpressed.

Across occurrences, exclude takes precedence over review, then allow. Unsafe sequence remains exclusion evidence even in an incomplete record. Allow means no exclusion from the assessed record under this policy. Exact matching does not validate an approximate similarity cutoff or patient-HLA decision. The historical Unsafe plus additional-forbidden assets retain their older policy identity.

Classification and sequence role answer different questions



Research parameters and coverage limits

Without usable matching protein staining, the upper RNA limits start at 4 for Vital, 8 for Protected and 16 for Manageable tissues, in the source's own TPM or nTPM units. Approved, Supported and Enhanced negative IHC respectively multiply the starting limit by 2, 4 and 8. Reviewed product-specific clinical evidence can add one doubling within the same cap. A value must be above the limit to trigger this RNA rule. Corneal CP10K uses separate limits, not a TPM conversion. These are policy choices, not measured safe exposures.

One compatible tumor sample above zero establishes the current cancer-presence criterion, not sufficient target abundance or cohort prevalence. Reproductive RNA uses a 5-unit criterion with assay-specific support requirements; reliable reproductive IHC and reviewed protein provide separate routes. Adult/placental diffuseness above 10% can block identity, never establish Unsafe.

The rare-cell panel uses a 1% detection screen with cell-support rules. Detection frequency is not abundance. Coverage is limited by sampled organs, donor age, activation states, paralog mapping and assay resolution. Optional panels remain disclosed when missing. A 95% Unsafe 9-mer overlap blocks practical candidate admission, but is not a clinical cross-reactivity validation.

The complete gate table below retains the actual executable rules and source IDs. Human thresholds are not transferred to dog or mouse; their native nominations remain unassessed until supported by a species-specific policy.

Does CTA mean expressed only in cancer?

No. CTA admission requires cancer expression plus qualifying reproductive or developmental association. The CTA total is not a count of proteins absent from all normal tissues, including testis, placenta, embryo and fetus. Passing normal-tissue thresholds is not a claim of zero normal expression.

The current data do support an observed-expression count: 40 cancer-expressed records have complete required bulk coverage and zero stored adult/reproductive RNA summaries. Of these, 22 have developmental measurements and 0 also have zero developmental RNA; 18 lack developmental measurements. Stored zeros are not proof of absent protein, and tiny positive RNA values are not validated protein production.

The separate low-expression screen finds 66 preliminary matches. Independent normal-cell evidence excludes 6 of them. A numerical profile match is not a final cancer-only designation; remaining matches retain explicit review or coverage requirements. The inventory audit lists every record, counterevidence source and measurement gap.

Cancer-specific altered sequences and peptides require a separate analysis from these reference-protein counts. A mutation, fusion or alternative transcript can create a target sequence even when the source gene is normally expressed; evidence for one such sequence cannot establish exclusivity of every product of the gene.

Reproduction and version changes

Selection policy: oncoref.cta-selection.2026-11-05.1. Role policy: oncoref.sequence-roles.2026-11-04.1. Translation policy: oncoref.translation-qualification.2026-11-04.1.

The 1.29.0 comparison remains pinned. The evidence and translation PRs also retain adjacent baselines, separating changes in category, subtype, role, assessment status and reasons. No target CTA count is an acceptance criterion.

Use the public decision, finding and sequence APIs. The final release receipt binds the decision exports, ontology, curation, qualification, bundle and generated figures. Complete and cached scientific replays must be byte-identical. Historical reports keep their original inputs and policies.

Presentation revision 2026-11-06.1 places the broader cancer-presence gate first after normal assessment and follows every registered nomination paper through the same cumulative gates. Classifications, thresholds and sequence roles are unchanged. Earlier release assets remain frozen.

Complete executable gate rules

Reproductive or developmental expression

Full Reproductive bulk ≥5 nTPM OR supported reproductive-cell RNA ≥5 nTPM (≥20 reads; indirect sites additionally need complete outside-cluster coverage and every outside value <5, with no curated normal-induction conflict) OR reliable mapped positive reproductive IHC for every member without an antibody conflict OR reviewed human reproductive protein covering every selected member OR supported developmental stage/lineage mean ≥5 TPM; ≥20 cells, ≥5 positive and ≥10% positive at ≥1 TPM. Presence evidence never overrides normal exclusions.

Sources: HPA v23 / GTEx consensus; HPA v23 normal-tissue IHC; HPA v23 tissue/cell clusters; Published human normal immune induction contexts; Tyser human gastrulation atlas; Petropoulos preimplantation atlas

Cancer expression

At least one measured compatible tumor sample >0 in native TPM or HPA pTPM OR reviewed reliable tumor protein. This establishes presence, not therapeutic abundance; do not add maxima from different samples.

Sources: OncoRef source-mapped tumor union; HPA v23 cancer sample measurements

HPA normal protein

Reliable non-Low positives in Vital/Protected supply Unsafe evidence; Manageable positives block CTA only; uncertain evidence remains unresolved. Low is flagged.

Sources: HPA v23 normal-tissue IHC

Required RNA coverage

Combined normal bulk (HPA consensus, HPA brain, GTEx v11) is the required RNA coverage panel: every nonreproductive tissue concept measured for every protein member. Missing coverage in the separate GTEx 52-context core, HPA tissue-cell, immune, DICE and cornea panels does not itself block admission; observed normal evidence from these panels still applies.

Sources: HPA v23 / GTEx consensus; HPA v23 human brain, 193 subregions; Direct GTEx v11 tissue medians; HPA v23 tissue/cell clusters; HPA v23 immune summaries; Monaco immune summaries via HPA v23

Reproductive concentration

HPA v23 average normal/(normal+Reproductive maximum) above 10% blocks the adult/placental route when its reference is positive and the comparison is complete. Use 38 contexts/29 anatomy units, including Disposable and excluding thymus; thymus remains Protected for hazard assessment. Supported embryonic evidence remains independent; its contrast is unassessed. This is not safety evidence.

Sources: HPA v23 / GTEx consensus

Translation qualification

Qualifying reference coding annotation or supported ncORF evidence is required for CTA admission, followed by a complete primary-locus annotated coding translation with no LoF/NMD annotation, incomplete tags, internal stops or non-AA20 residues; the selected common longest sequence must qualify for every member. All candidate records still undergo normal-evidence assessment. Translation failure requires review, never Unsafe.

Sources: Ensembl 115 coding translations

Normal bulk

One inclusive tissue/IHC ceiling: above it in Vital/Protected supplies Unsafe evidence; above it in Manageable blocks CTA, and above 128 (the top Manageable ceiling) supplies Unsafe evidence. No second RNA boundary.

Sources: HPA v23 / GTEx consensus; HPA v23 human brain, 193 subregions; Direct GTEx v11 tissue medians

HPA tissue clusters

RNA requires ≥20 reads and uses the single ceiling. IHC credit requires an exact or declared compartment match; unrelated cells in the same tissue cannot inherit it.

Sources: HPA v23 tissue/cell clusters

HPA immune

Central cell-specific tiers with native nTPM limits and no matched negative-IHC credit; reproductive/placental location remains neutral.

Sources: HPA v23 immune summaries

Monaco immune

Central cell-specific tiers with native TPM limits and no matched negative-IHC credit.

Sources: Monaco immune summaries via HPA v23

DICE immune

Third-highest reported TPM value; central cell-specific tiers. Unpaired group distributions retain bounds. Missing coverage alone does not exclude.

Sources: DICE immune distributions

Adult cornea

Recurrent native CP10K: ≥5 positive cells and ≥20 counts in ≥2 sample groups. Nominal RNA limits /100; not a TPM conversion. Missing coverage alone does not exclude.

Sources: Collin adult corneal atlas

Accepted normal-protein evidence

Accepted positive protein findings exclude unless extracted Low intensity retains a flag; unquantified does not mean negative.

Sources: Reviewed normal protein and clinical exclusions

Reviewed normal-cell knowledge

Reviewed adult human normal protein, or localized RNA with orthogonal normal-cell function, in Vital/Protected supplies Unsafe evidence. Sparse RNA, uncertain species/stage or unresolved identity produces a review hold. Activation and rare-cell frequency do not exempt normal cells. Every finding retains its source and limitations.

Sources: Reviewed human gene knowledge: normal cells and reproductive protein

Clinical safety exclusion

Retain reviewed product-specific clinical injury counterevidence and causal-attribution limits; no admission shortcut from clinical use.

Sources: Reviewed normal protein and clinical exclusions

Rare vital-tissue cell expression

At least 1% detection in a centrally assigned Vital population in the existing brain/spinal-cord/heart/lung CELLxGENE panel supplies Unsafe evidence. At least 3,000 cells and 10 expressing cells are required. Excluded or unmeasured populations are coverage limits, not allowances. Detection frequency is distinct from abundance.

Sources: CELLxGENE rare-cell screen

Rare Protected-cell expression review

At least 1% detection in a centrally assigned Protected population in the existing CELLxGENE panel blocks CTA admission without itself establishing Unsafe. The same cell-support criteria apply. Source age restrictions and detection artifacts remain limitations.

Sources: CELLxGENE rare-cell screen

External validated normal protein

Use categorical/quantitative staining only within verified reagent, tissue and cell scope; no inherited HPA reliability label.

Sources: Curated external staining observations

Unsafe sequence containment

A would-be CTA with at least 95% of its 9-mers also present in Unsafe proteins becomes Other:Contained: Unsafe sequence is blacklisted, so almost no peptide survives. Computed over all protein-coding transcripts of Ensembl 115. This is a sequence rule, not expression evidence.

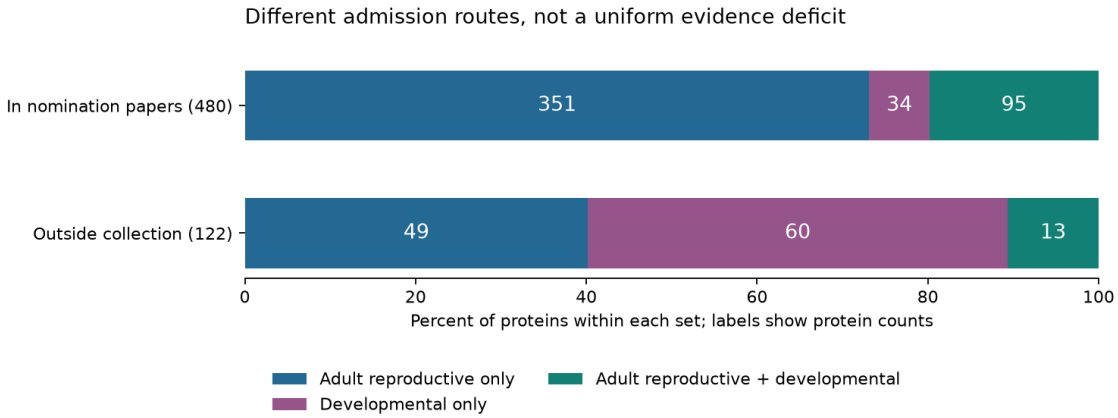
Sources: Ensembl 115 coding translations; Sequence overlap with released Unsafe proteins

CTA and Other assignments

CTA requires all applicable admission gates, required coverage and resolved normal evidence. Unsafe exclusion takes precedence. Other includes assessed noncandidates, held/incomplete evidence, nested microproteins and sequence-contained candidates. Subtype and assessment_status answer different questions; an identity failure cannot erase a normal-evidence hold. Sequence roles require that independent status.

Sources: All released decision gates

The 122 CTAs outside the paper collection



The 602 released CTAs include 480 proteins in the 15-paper nomination union and 122 outside it (123 loci). Absence from this finite collection does not establish novelty or absence from the literature.

Of the 122, 60 qualify through developmental evidence alone, 49 through adult reproductive evidence alone and 13 through both. Only 34 of the 480 registered proteins use the developmental-only route. These are admission routes in the measured datasets, not exclusive biological identities.

Median highest tumor RNA is 35.6 versus 33.5 TPM. This does not suggest uniformly weaker tumor maxima, but maxima do not measure prevalence, replication or specificity. All 122 pass the annotation-based translation rule; none has a released gene-knowledge finding. An empty ledger is not a negative literature review.

Complete 122-protein table, route comparison, linked primary-literature findings and follow-up priorities

Focused primary-literature review of representative mechanisms and possible conflicts among the 122 released CTAs absent from the 15-paper nomination union. This is not an exhaustive literature search for all 122. Findings here are report annotations awaiting structured curation; they do not modify release 1.31.0 classifications or establish receptor recognition.

TBXT - Missing cancer-target literature

Brachyury was already a clinical cancer-vaccine target. Its absence from this nomination union is a collection gap, not evidence of novelty.

Phase I Trial of a Yeast-Based Therapeutic Cancer Vaccine (GI-6301) Targeting the Transcription Factor Brachyury

Trial record

Scope: Human clinical vaccine study. Limits: Vaccine immunogenicity or tolerability does not establish safety of a particular engineered TCR, CAR or peptide. TBXT is also a developmental lineage regulator.

SEMG2 - Missing explicit CTA literature

The authors explicitly describe SEMG2 as a cancer-testis antigen and report tumor RNA and cancer-cell-line protein evidence.

SEMG1/2 augment energy metabolism of tumor cells

Scope: Human tumor data and cancer cell lines. Limits: The label does not establish absence from every normal tissue, nor peptide presentation or clinical safety. Do not automatically extend SEMG1 findings to SEMG2.

SEMG2 - Normal-tissue conflict requiring curation

The study reports SgII transcripts outside reproductive organs, including trachea and kidney, and airway basal-cell staining with semenogelin-reactive antibodies. SEMG2 can therefore be both a published CTA nomination and a protein with normal-tissue counterevidence.

Semenogelin I and II, the predominant human seminal plasma proteins, are also expressed in non-genital tissues

Scope: Human tissue RT-PCR and immunohistochemistry. Limits: The antibodies recognize both SgI and SgII, so the staining cannot be assigned exclusively to SEMG2. RNA, mixed-reactivity staining and exact protein attribution must be assessed separately.

PATE1 - Missing cancer/reproductive literature

The original PATE study explicitly joins cancer expression with normal prostate and testis expression. This is a reproductive-organ candidate, not necessarily a male-germ-cell-specific CTA.

PATE, a gene expressed in prostate cancer, normal prostate, and testis, identified by a functional genomic approach

Scope: Human tissue expression and functional genomics. Limits: Resolve the historical PATE name to the exact present locus and translation before adding membership. Related PATE-family studies cannot be assigned indiscriminately to PATE1.

LEUTX, DPRX - Early embryo regulation

LEUTX and DPRX participate in an early embryo transcriptional program. The paper also examines limited cancer-cell-line expression; it does not establish broad cancer restriction.

The human PRD-like homeobox gene LEUTX has a central role in embryo genome activation

Scope: Human embryos and human embryonic stem-cell overexpression. Limits: Overexpression models and embryonic expression do not prove endogenous tumor protein. The reported cancer examples and disease-state expression require isoform-aware interpretation.

DPRX, DUXA, DUXB, TPRX2 - Early embryo regulation

These genes are represented in a primary study of preimplantation homeobox factors. They belong to a literature organized around embryonic genome activation, rather than classical testis-biased cancer lists.

Characterization and target genes of nine human PRD-like homeobox domain genes expressed exclusively in early embryos

Scope: Human embryo transcripts and human stem-cell functional assays. Limits: The paper's expression scope is finite; its title is not proof of universal somatic absence. Transcript and construct function are not tumor antigen presentation.

LEUTX, ZNF280A, TPRX2, DUXB - Early embryo regulation

LEUTX regulates an early-development program including ZNF280A and TPRX2. The study highlights TPRX2's historical pseudogene annotation despite transcriptional and functional evidence.

Comprehensive characterization of the embryonic factor LEUTX

Scope: Human cell models, proteomics and transcriptional assays. Limits: A response to experimentally induced LEUTX is not evidence that every downstream protein is naturally translated in tumors. Historical annotation changes may explain some list omissions, but do not prove why each source omitted a gene.

LEUTX - Fusion-specific cancer evidence

A YAP1::LEUTX fusion provides cancer relevance for LEUTX, in a specific molecular form.

Whole genome profiling of rare pediatric thoracic tumors elucidates a YAP1::LEUTX fusion in an unclassified biphasic embryonal neoplasm

Scope: Human tumor case study. Limits: A fusion does not demonstrate expression or translation of the full-length reference LEUTX protein used by this release.

WEE2 - Oocyte and maternal-effect biology

Human oocyte WEE2 has direct protein and functional support, including loss-of-function-associated fertilization failure and experimental rescue.

Homozygous Mutations in WEE2 Cause Fertilization Failure and Female Infertility

Scope: Human genetics, affected oocytes and rescue experiments. Limits: This is strong reproductive-function evidence, not tumor translation or safety evidence. A developmental-only admission route in the current datasets does not mean the protein has no adult oocyte biology.

KHDC3L - Oocyte and maternal-effect biology

KHDC3L/ECAT1 was identified in human oocytes, supporting a female reproductive protein association.

Identification of Maturation-Specific Proteins by Single-Cell Proteomics of Human Oocytes

Scope: Human oocyte proteomics. Limits: Protein identification in oocytes does not establish tumor protein or absence elsewhere. Historical ECAT1/C6orf221 aliases must retain exact identity.

NLRP14 - Oocyte and maternal-effect biology

NLRP14 has a dedicated female-fertility and early-development literature that a testis-centered nomination collection may underrepresent.

NLRP14 deficiency causes female infertility with oocyte maturation defects and early embryonic arrest by impairing cytoplasmic UHRF1 abundance

Scope: Human infertility investigation and mechanistic experiments. Limits: Separate human observations from experimental model results when curating; this paper alone does not qualify a tumor antigen.

PABPN1L - Oocyte and maternal-effect biology

Pabpn1l knockout disrupts early embryonic development in mice, offering a mechanistic explanation for a maternal-to-zygotic expression pattern.

PABPN1L mediates cytoplasmic mRNA decay as a placeholder during the maternal-to-zygotic transition

Scope: Mouse oocytes and embryos. Limits: Animal function cannot establish the human normal-tissue distribution or tumor translation. PABPN1L must not be conflated with PABPC1L.

PATE1, WFDC8 - Reproductive tract secretion

The study identifies epididymal enrichment for PATE1 and WFDC8. Male reproductive support includes somatic secretory compartments as well as germ cells.

In-depth mapping of human testicular and epididymal proteins and their functional association with spermatozoa

Scope: Human reproductive tissues and proteins. Limits: Sperm association does not imply synthesis by sperm; reproductive expression is not cancer specificity.

RNASE9 - Reproductive tract secretion

Human epididymal RNASE9 was cloned and localized, with sperm-associated protein evidence.

Cloning, expression and location of RNase9 in human epididymis

Scope: Human tissue RNA and immunolocalization. Limits: A finite tissue panel cannot establish absence everywhere else; binding to sperm is distinct from germ-cell production.

MRGPRX1 - Normal-tissue conflict requiring curation

The human experiments report MRGPRX1 RNA in donor sensory ganglia, single-neuron detection and agonist-responsive neurons. This is a concrete normal sensory-neuron concern outside the current brain/heart/lung rare-cell panel.

Activation of pruritogenic TGR5, MrgprA3, and MrgprC11 on colon-innervating afferents induces visceral hypersensitivity

Scope: Mouse mechanisms plus human dorsal-root-ganglion expression and functional assays. Limits: The study is not a receptor-specific T-cell toxicity experiment. Exact human cell attribution, functional specificity and policy disposition require structured review before changing the released call.

SP7 - Normal-tissue conflict requiring curation

Human SP7/osterix has normal osteoblast-expression evidence. Bone-forming cells deserve specific evaluation rather than an inference from low routine bulk RNA.

Expression of alternatively spliced isoforms of human Sp7 in osteoblast-like cells

Scope: Human primary osteoblast cultures, including adult cells. Limits: Cultured-cell expression is not a quantified in-vivo abundance or immune-injury threshold; isoforms and differentiation state matter.

KLF14 - Normal somatic function requiring review

KLF14 has established normal human adipose expression and metabolic function; it should not be described as reproductive-only.

Regulatory variants at KLF14 influence type 2 diabetes risk via a female-specific effect on adipocyte size and body composition

Scope: Human adipose cohorts and primary cells, with mouse experiments. Limits: The registry treats adipose as a conditional Manageable compartment. Normal function alone does not automatically imply Unsafe under that policy; abundance, exact cell scope and compensation must be reviewed.

CALHM3 - Sensory-function follow-up

CALHM3 has a defined taste-signaling function, identifying a sensory compartment to investigate.

CALHM3 Is Essential for Rapid Ion Channel-Mediated Purinergic Neurotransmission of GPCR-Mediated Tastes

Scope: Mouse taste-cell genetics and heterologous channel experiments. Limits: Mouse taste-cell function and human constructs do not by themselves establish adult human tissue localization or immune-mediated toxicity.

BTN1A1 - Physiological-state coverage gap

The current HPA page reports High protein expression in lactating glandular cells, illustrating normal expression conditional on reproductive physiology.

Human Protein Atlas: BTN1A1 staining in lactating breast

Scope: Current human tissue immunohistochemistry. Limits: This live page is not the pinned HPA v23 input. Reagent reliability, version differences and the precise physiological compartment need review; lactation is not equivalent to fetal expression.

Plain-language sequence decisions

No exclusion recorded (allow): the assessed record has no policy exclusion; this is not a target recommendation. Unresolved (review): do not automatically clear the sequence until the relevant questions are answered. Exclude: retain as exclusion evidence; this takes precedence over review and allow.

Missing measurements: Required normal tissues or member loci have missing observations.

Conflicting or uncertain normal evidence: A normal signal, reagent result or causal interpretation remains unresolved.

Uncertain identity: The gene, locus, transcript or source mapping is not resolved.

Uncertain translation: Coding support or translation completeness does not qualify.

Ambiguous sequence: Stops, ambiguous residues or non-AA20 characters prevent automatic exact-record use.

Not assessed: The record or isoform lacks an applicable decision; the selected representative cannot clear every isoform.

RNA-limit table and full decision explanations

Data-source guide

Measurement datasets, literature nominations, reviewed findings and derived rules are distinct. Blank source cells in earlier reports did not mean evidence-free decisions: rare-cell rules use CELLxGENE, and sequence containment uses Ensembl translations plus current Unsafe assignments.

Reviewed human gene knowledge: normal cells and reproductive protein

Measurement: source-native qualitative and quantitative observations. Scope: explicit per-finding stage and activation state. Exact gene/protein attribution; sparse RNA and unresolved species/isoforms remain review holds. Search leads are not accepted findings.

Durante et al. 2020 adult human olfactory atlas supplementary tables

Measurement: positive neurons; cluster detection fraction. Scope: adult. Exact reported genes enter knowledge review holds; sparse detection is not TPM, protein, donor replication or proof of absence for unlisted genes.

Verbeurgt et al. 2014 whole human olfactory mucosa

Measurement: native qPCR Cq; assay IDs and donor identity. Scope: adult; 26 donor samples. OR10A4 normal RNA supplies a reviewed hold. Native values are not TPM, protein or a clinical toxicity estimate; unlisted genes are not measured zero.

Verbeurgt et al. 2014 whole human olfactory mucosa

Measurement: estimated RNA copies per 20 ng total RNA. Scope: adult; 26 donor samples. OR10A4 normal RNA supplies a reviewed hold. Native values are not TPM, protein or a clinical toxicity estimate; unlisted genes are not measured zero.

Published human normal immune induction contexts

Measurement: qualitative reported RNA induction. Scope: adult; activated. Qualitative counterevidence to indirect reproductive-site restriction; no TPM or clinical-toxicity inference

HPA v23 / GTEx consensus

Measurement: nTPM. Scope: adult; placenta handled separately. Bulk source summaries; not exhaustive pediatric, pregnancy or rare-cell coverage

Tyser human gastrulation atlas

Measurement: TPM. Scope: Carnegie stage 7. One embryo; cell support does not establish independent-donor replication

Petropoulos preimplantation atlas

Measurement: TPM. Scope: 3–7 days, stage × inferred lineage. 1,519 measured cells; stage and embryo counts retained; not coverage of all fetal stages

HPA v23 human brain, 193 subregions

Measurement: nTPM. Scope: adult. Maximum subregion mean after within-context group sums; no automatic negative-IHC credit

Direct GTEx v11 tissue medians

Measurement: TPM. Scope: adult. 52 native whole-organ contexts; 14 microdissected and two cultured/transformed contexts remain outside this adapter. Exact whole-organ HPA negative-IHC grades apply; donor medians do not exclude donor outliers. Native reader preserves all 68 contexts.

OncoRef source-mapped tumor union

Measurement: TPM. Scope: cohort-specific. 132 compatible cohorts; source namespaces deduplicate views, not patients across studies

HPA v23 cancer sample measurements

Measurement: pTPM. Scope: cohort-specific. Protein-coding RNA denominator; no numerical protein or peptide abundance

HPA v23 tissue/cell clusters

Measurement: nTPM. Scope: source-labelled adult; reproductive locations explicit. At least 20 reads; missing cell types and disease states remain uncovered

HPA v23 immune summaries

Measurement: nTPM. Scope: adult. Sorted populations retain native units; no matched negative-IHC relaxation

Monaco immune summaries via HPA v23

Measurement: TPM. Scope: adult. Sorted populations retain native units; no matched negative-IHC relaxation

DICE immune distributions

Measurement: TPM. Scope: adult. 15 distributions; third-highest value; unpaired locus distributions retain bounds; optional coverage

Collin adult corneal atlas

Measurement: CP10K. Scope: adult. Two supported sample groups; optional coverage; CP10K is not length-normalized TPM

HPA v23 normal-tissue IHC

Measurement: HPA categorical level and reliability. Scope: adult. Reliability describes evidence support, not a calibrated false-negative probability

Curated external staining observations

Measurement: native categorical findings. Scope: source-specific. Reagent, tissue, cell type and validation scope remain explicit; no inherited HPA grade

Reviewed normal protein and clinical exclusions

Measurement: categorical reviewed findings. Scope: source-specific. Qualitative reported Low is distinct from HPA Low and numerical intensity; clinical evidence is product-specific

Letetresgene Autoleucel in Advanced/Metastatic Myxoid/Round Cell Liposarcoma

Measurement: reviewed clinical attribution. Scope: adult clinical cohort. Peptide/HLA/receptor scope; targeted review, not all trials; no species transfer

Autologous T cell therapy for PRAME+ advanced solid tumors in HLA-A*02+ patients: a phase 1 trial

Measurement: reviewed clinical attribution. Scope: adult clinical cohort. Peptide/HLA/receptor scope; targeted review, not all trials; no species transfer

Autologous T cell therapy for MAGE-A4+ solid cancers in HLA-A*02+ patients: a phase 1 trial

Measurement: reviewed clinical attribution. Scope: adult clinical cohort. Peptide/HLA/receptor scope; targeted review, not all trials; no species transfer

SPEARHEAD-1: a single-arm phase 2 trial of afamitresgene autoleucel in advanced synovial sarcoma and myxoid/round cell liposarcoma

Measurement: reviewed clinical attribution. Scope: adult clinical cohort. Peptide/HLA/receptor scope; targeted review, not all trials; no species transfer

Identification of a Titin-Derived HLA-A1-Presented Peptide as a Cross-Reactive Target for Engineered MAGE A3-Directed T Cells

Measurement: reviewed clinical attribution. Scope: adult clinical cohort. Peptide/HLA/receptor scope; targeted review, not all trials; no species transfer

Guillain-Barre syndrome observed with adoptive transfer of lymphocytes genetically engineered with an NY-ESO-1 reactive T-cell receptor

Measurement: reviewed clinical attribution. Scope: adult clinical cohort. Peptide/HLA/receptor scope; targeted review, not all trials; no species transfer

NCT03709706 protocol: NY-ESO-1/LAGE-1A shared peptide and c259 receptor

Measurement: reviewed clinical attribution. Scope: adult clinical cohort. Peptide/HLA/receptor scope; targeted review, not all trials; no species transfer

NCT03132922 protocol: MAGE-A4 c1032 target peptide

Measurement: reviewed clinical attribution. Scope: adult clinical cohort. Peptide/HLA/receptor scope; targeted review, not all trials; no species transfer

First-in-human dose escalation trial to evaluate the clinical safety and efficacy of an anti-MAGEA1 autologous TCR-transgenic T cell therapy in relapsed and refractory solid tumors

Measurement: reviewed clinical attribution. Scope: adult clinical cohort. Peptide/HLA/receptor scope; targeted review, not all trials; no species transfer

Phase I clinical trial evaluating the safety and efficacy of ADP-A2M10 SPEAR T cells in patients with MAGE-A10+ advanced non-small cell lung cancer

Measurement: reviewed clinical attribution. Scope: adult clinical cohort. Peptide/HLA/receptor scope; targeted review, not all trials; no species transfer

NY-ESO-1-specific redirected T cells with endogenous TCR knockdown mediate tumor response and cytokine release syndrome

Measurement: reviewed clinical attribution. Scope: adult clinical cohort. Peptide/HLA/receptor scope; targeted review, not all trials; no species transfer

Cancer regression and neurological toxicity following anti-MAGE-A3 TCR gene therapy

Measurement: reviewed clinical attribution. Scope: adult clinical cohort. Peptide/HLA/receptor scope; targeted review, not all trials; no species transfer

Re-examination of MAGE-A3 as a T-cell Therapeutic Target

Measurement: reviewed clinical attribution. Scope: adult clinical cohort. Peptide/HLA/receptor scope; targeted review, not all trials; no species transfer

CELLxGENE rare-cell screen

Measurement: % cells with detected RNA. Scope: source-specific. Existing brain/spinal-cord, heart and lung panel only; >=3,000 cells and >=10 positive cells. >=1% detection is a policy trigger, not an abundance measurement or proof of receptor recognition.

Ensembl 115 coding translations

Measurement: annotation / amino-acid sequence. Scope: source-specific. Qualifying annotated CDS is accepted translation support; it does not prove protein detection in a tumor. Exact selected sequence and every member locus must qualify.

Sequence overlap with released Unsafe proteins

Measurement: % shared 9-mers. Scope: source-specific. Computed from Ensembl 115 coding translations and current Unsafe assignments. >=95% containment blocks CTA admission; it does not itself establish tissue expression.

All released decision gates

Measurement: classification / reasons. Scope: source-specific. Derived result, not an independent experimental source. Unsafe takes precedence over CTA admission; Other retains its evidence status.

Worked decisions and primary sources

PRAME - CTA

screened; sequence role: allow. Reasons: All declared gates pass.

Translation: Qualified selected primary-locus record.

unresolved / context: The historical blanket primary-stain hold is superseded by the current HPA level/reliability and compartment rules, including prior reagent eligibility. Original raw staining and unresolved specificity remain visible; this policy revision is not new validation of the reagent or evidence of absence. No second historical IHC veto is applied.

HPA v23 primary IHC; repeated family profiles are not independent studies

All mapped positive rows, profile notes and reagent summaries reviewed; no independent histology re-read or new experiment. Reliable gene-profile labels do not validate each raw stain. RNA/protein donors are unpaired. Retain observed intensities and identity ambiguity; the decision is a conservative research hold.

accepted / context: Reported Low intensity remains a flag without an automatic protein veto under the current policy; retain the observation and source-specific limitations.

33890816; 35973038

Primary 2021 abstract; 2022 normal-tissue methods/results inspected via primary article search excerpts. Full 2021 methods unavailable; abstract-level conservative acceptance, not proven therapeutic injury. IMA203 evidence concerns a selected peptide/HLA/TCR product and is preserved separately.

MAGEC2 - Other:Held

normal_evidence_hold; sequence role: review. Reasons: Reviewed normal-cell knowledge

Translation: Qualified selected primary-locus record.

unresolved / review: Material normal-tissue finding remains unresolved; block CTA admission and require sequence review without establishing Unsafe. Negative bulk profiles do not resolve assay identity.

HCA587 antigen expression in normal tissues and cancers: correlation with tumor differentiation in hepatocellular carcinoma

Primary abstract rechecked; publisher full text unavailable in this review. Original tissue and antibody controls remain unverified.. Abstract does not settle reagent specificity; no demonstrated target-mediated clinical injury. A neural protein conflict warrants a hold, but the historical antibody result does not establish target-mediated clinical injury or resolve MAGE-family cross-reactivity.

LGALS13 - Other:Held

normal_evidence_hold; sequence role: review. Reasons: Reviewed normal-cell knowledge

Translation: Qualified selected primary-locus record.

unresolved / review: Material normal-tissue finding remains unresolved; block CTA admission and require sequence review without establishing Unsafe. Negative bulk profiles do not resolve assay identity.

Isolation and sequence analysis of a cDNA encoding human placental tissue protein 13 (PP13)

Primary abstracts rechecked (PMIDs 10527825 and 19497882); original adult/fetal specimen labels, bands and reagent controls not independently re-inspected.. Adult versus fetal samples and original protein bands require full-paper adjudication.

Conflicting assay identity is unresolved; neither a proven unsafe classification nor unconditional inclusion is justified by these abstracts alone.

PLCZ1 - Other:Held

normal_evidence_hold; sequence role: review. Reasons: Reviewed normal-cell knowledge

Translation: Qualified selected primary-locus record.

rejected / context: Preserve rejection of the PLCZ1 protein interpretation: the immunostaining measured CFTR. The separately retained qualitative pancreatic RNA finding remains unresolved.

Rescuing defective pancreatic secretion in cystic fibrosis by suppression of a newly identified gene

Primary full text: Methods, normal-tissue RT-PCR/Figure 1, and CFTR immunolocalisation. Original isoform mapping and quantitative expression remain unresolved.. Human pancreatic RNA is qualitative, not nTPM. Disease/cell-model functional experiments and mouse measurements do not establish healthy human protein abundance.

unresolved / review: Material normal-tissue finding remains unresolved; block CTA admission and require sequence review without establishing Unsafe. Negative bulk profiles do not resolve assay identity.

Rescuing defective pancreatic secretion in cystic fibrosis by suppression of a newly identified gene

Primary full text: Methods, normal-tissue RT-PCR/Figure 1, and CFTR immunolocalisation. Original isoform mapping and quantitative expression remain unresolved.. Human pancreatic RNA is qualitative, not nTPM. Disease/cell-model functional experiments and mouse measurements do not establish healthy human protein abundance.

MAGEA12 - Unsafe

screened; sequence role: exclude. Reasons: Rare vital-tissue cell expression

Translation: Qualified selected primary-locus record.

SERPINA2 - Other:Translation

translation_review; sequence role: review. Reasons: Translation qualification

Translation: incomplete_annotation_tag | incomplete_start | incomplete_stop | noncanonical_start_residue | nonqualifying_transcript_biotype

MROH5 - Unsafe

screened; sequence role: exclude. Reasons: Translation qualification; Rare vital-tissue cell expression

Translation: ambiguous_or_interrupted_sequence | nonqualifying_transcript_biotype

SELENOV - Other:Translation

translation_review; sequence role: review. Reasons: Translation qualification

Translation: ambiguous_or_interrupted_sequence